

Insects contribute considerably to global crop pollination, [with pollination deficits being](#) documented for multiple entomophilous or pollinator-dependent crops. Different cultivars of crops are being cultivated within and across production regions, so it is essential to understand the cultivar variability of pollination deficits. Here, we used a dataset from 286 sites from multiple production regions to develop a synthesis on pollination deficits in two widely cultivated highbush blueberry cultivars, ‘Bluecrop’ and ‘Duke’. Additionally, we determined if bee visitation or bee richness reduces pollination deficits in these cultivars. On average, neither cultivar showed pollination deficits regarding fruit set. However, for ‘Bluecrop’ we found pollination deficits for berry weight and seed set, which was not the case for ‘Duke’. Increasing total bee visitation reduced pollination deficits of both berry weight and seed set for ‘Bluecrop’. More specifically, a non-linear, negative exponential model best predicted this relation between bee visitation and pollination deficits. Our results highlight that pollination deficits and responses to pollinator visitation are variable between different cultivars of a single crop, which suggests opportunities to use certain cultivars that are less dependent on insect-mediated pollination in landscapes and regions where pollination services have been compromised. In addition, the non-linear response between bee visitation and pollination deficits suggests that optimal bee visitation rates need to be determined to improve pollination management and crop yield and to support accurate economic valuations of pollination services.

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Crop production

highbush blueberry

Vaccinium spp.

seed set

fruit weight

fruit set

LICENCE



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